

ogy. It has a built-in amplifier to support Class 1 transmit power. This chip features a packet loss concealment technology that reduces packet loss and bit error rates. This makes it a good choice for audio output devices that rely on a Bluetooth link for audio streaming.

One of the lowest-voltage Bluetooth Smart controllers currently available is the EM9301 by EM Microelectronic. This controller operates at 0.8 V, which makes it a good choice for cases where power might be a very precious resource—such as in solar cells, piezoelectric and other energy harvesters.

Multifunction processors. Unlike single-chip processors, these RF silicon chips come with a variety of communication technologies apart from Bluetooth. The chips are targeted at applications where a variety of wireless technologies would be used simultaneously. As such, they come with technology that works to mitigate potential interference between the various wireless technologies (explained later).

Broadcom's BCM2076 is one such combo chip that features Bluetooth Smart and FM, and also supports both the GPS constellation as well as the GLONASS constellation. There is also an integrated technology in the chip, called InConcert, that allows Bluetooth and Wi-Fi signals to coexist without interfering.

BLE system-on-chips. Since the proliferation of the Internet of Things,

Power profiles for BLE devices

- **Health Thermometer Profile (HTP):** Used for medical temperature measurement
- **Blood Pressure Profile (BLP):** Used in blood pressure monitoring applications
- **Glucose Profile (GLP):** Used in blood glucose meters
- **Heart Rate Profile (HRP):** Intended for use in heart rate measurement devices
- **Cycling Speed and Cadence Profile (CSCP):** Runs on a device integrated to a bike, monitoring its cadence and wheel speed
- **Find Me Profile (FMP):** A trigger on one device sends an alert to another. The former is known as the target, while the latter becomes the locator.
- **Proximity Profile (PPX):** Used in applications that detect whether another device is within close physical range
- **Phone Alert Status Profile (PASP):** It allows a device to receive notifications from another device and trigger an action

or the Electronics of Things as we may like to call it, there has been a surge in the demand to create wireless personal area networks for each tech-savvy person (including me). We have most of our personal belongings tagged with a Bluetooth 'sticker,' which gives these objects a virtual representation of themselves. The benefit is to track and easily locate every piece of accessory that I carry about with me. While single-chip Bluetooth radios are good, cases like these demand a dedicated system-on-chip (SoC) that can handle all the functions on one very small chip.

One of the SoC series that is now famous is the nRF51 series from Nordic Semiconductor. These are highly flexible, multi-protocol, SoC devices for ultra-low-power wireless applications. They run on an ARM Cortex M0 processor core, and the memory and feature specs differ amongst the differ-

ent variants of the chip.

One benefit of these SoCs is that different devices are compatible across product families as long as they are based on the nRF51 with similar general-purpose input/output (GPIO) count. A major benefit that designers would see is "freedom from proprietary software frameworks," as Nordic puts it.

How to make the design process easier

While using the aforementioned modules could be either difficult or time-consuming for integrators, many solution providers have started working on making them pluggable type by integrating them with other peripherals and control boards.

"A few shield designs are already available for Arduino, although these are costly for small/one-time users. Many of the solution providers who have come up with integration modules have also come up with supporting libraries and tools to test or simulate them. Arduino too has a library for Bluetooth use. We are making further efforts to improve existing Bluetooth libraries and bring them to an abstraction level where users would be able to match time-to-market easily and deliver before time," explains T. Anand, MD, Knewron.

Vijayasathya Shastri, application engineer at Texas Instruments (India), explains TI's efforts to make the design process easier using its products: "First, meeting the most predominant design challenge, TI offers a royalty-free certified stack that addresses 80 per cent of the profiles listed on BT SIG. For those without any prior RF designing experi-

Recommended BLE modules

There are many chipset and module makers, viz, Broadcom, ConnectBlue, Panasonic, Bluegiga, InsightSiP and Blue Radio. All of their modules are based on common TI CC2540/41 chipset stacks. Only InsightSiP's product is built with Nordic nRF8001 chipset.

External host interface. In Blue Radio module, the built-in API for external host uses AT commands, which are extremely easy to integrate with existing libraries. Bluegiga has a binary API dependant on ANSI C library.

Application hosting. In case of BLE, to save on power, cost and complexity, applications are typically hosted on the modules themselves. Application hosting on Bluegiga module is free, while it costs almost 60 per cent less on Blue Radio compared to other modules.

Pricing. Blue Radio module is costlier, while Bluegiga module is the cheapest and the rest of them are averagely priced.

Certifications. Modules based on a technology like Bluetooth must be certified for CE and FCC standards. Other certifications are not mandatory but desirable. Bluegiga modules carry five certifications compared to basic certifications carried by other modules. Having certifications already in place certainly reduces cost and speeds up the development.

All in all, it seems that Blue Radio and Bluegiga modules should be preferred in designs followed by Panasonic and ConnectBlue modules.

— T. Anand, MD, Knewron